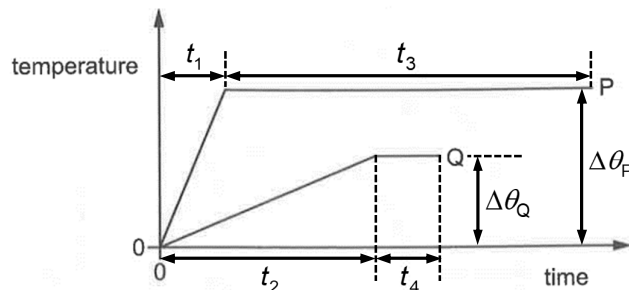


12

C

For solids P and Q of the same masses m , heated at the same rate P .



Before melting,

$$P: mc_P \Delta\theta_P = Pt_1 \quad \text{--- (1)}$$

$$Q: mc_Q \Delta\theta_Q = Pt_2 \quad \text{--- (2)}$$

$$\frac{(1)}{(2)} : \frac{c_P \Delta\theta_P}{c_Q \Delta\theta_Q} = \frac{t_1}{t_2}$$

$$\frac{c_P}{c_Q} = \frac{t_1}{t_2} \times \frac{\Delta\theta_Q}{\Delta\theta_P}$$

$$\text{Since } \frac{t_1}{t_2} < 1, \frac{\Delta\theta_Q}{\Delta\theta_P} < 1, \frac{c_P}{c_Q} < 1, c_P < c_Q$$

During melting,

$$P: ml_P = Pt_3 \quad \text{--- (3)}$$

$$Q: ml_Q = Pt_4 \quad \text{--- (4)}$$

Qn	Answer	Solution
		$\frac{(3)}{(4)} : \frac{l_P}{l_Q} = \frac{t_3}{t_4}$ Since $\frac{t_3}{t_4} > 1$, $\frac{l_P}{l_Q} > 1$, $l_P > l_Q$
13	B	Maximum kinetic energy of the air molecule